

Literature Answer to the L -Orthogonal Sequence Question

Agent lane 17

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Status

This is a lightweight `literature_already_answered` packet. The remaining headline question from Aviles–Martinez–Cervantes–Rueda [1] is explicitly answered by the later paper of Aviles–Martinez–Cervantes–Poveda–Saenz [2].

Original Question

In [1, Question 1.1], the authors ask:

Let X be a Banach space with an L -orthogonal sequence $\{x_n\}$. Is there an L -orthogonal element u in X^{**} ?

They also ask the stronger weak-star-closure variant in [1, Question 1.2]. The 2021 paper proves positive consistency results and shows that the weak-star-closure variant is undecidable in ZFC, but it leaves open whether a negative answer to Question 1.1 is consistent.

Answering Paper

The later paper [2] explicitly identifies this problem. Its abstract says that it provides a definitive answer to [1, Question 1.1]. In [2, Main Theorem, “Independence of L -elements”], the authors prove:

The existence of a Banach space with an L -orthogonal sequence but without L -orthogonal elements in its bidual is independent of ZFC.

Thus the original Question 1.1 is settled in the precise set-theoretic sense: the positive universal statement holds in some models of ZFC, while in other models there is a Banach space counterexample.

Scope and Duplicate Check

This packet does not claim a new proof. It records an explicit later-literature answer. A local registry hit for arXiv:2501.06537 already existed, but it concerns the later paper’s quantitative Question 2 about epsilon-strong Q -measures, not the headline answer to arXiv:2104.05535. Web search on 2026-06-20 for exact phrases around “2104.05535”, “ L -orthogonal sequences”, and “without L -orthogonal elements” located arXiv:2501.06537 as the decisive supporting source.

References

- [1] A. Aviles, G. Martinez-Cervantes, and A. Rueda Zoca, *L-orthogonal elements and L-orthogonal sequences*, arXiv:2104.05535, 2021. <https://arxiv.org/abs/2104.05535>
- [2] A. Aviles, G. Martinez-Cervantes, A. Poveda, and L. Saenz, *A Banach space with L-orthogonal sequences but without L-orthogonal elements*, arXiv:2501.06537, 2025. <https://arxiv.org/abs/2501.06537>